RAG-Based Customer Support Assistant - Low Level Design

Module-Level Design

• **Document Processing Module:** Utilizes PyPDFLoader to parse raw PDF data into structured text objects.

• **Chunking Module:** Implements RecursiveCharacterTextSplitter to break documents into segments of 500 characters with 50-character overlaps.

• **Embedding Module:** Employs the all-MiniLM-L6-v2 model to transform text chunks into 384-dimensional vector representations.

• **Vector Storage Module:** Manages local persistence and indexing of vectors using ChromaDB.

• **Retrieval Module**: Executes similarity searches to fetch the top 3 most relevant context chunks for any given query.

• **Query Processing Module**: Interacts with the Llama 3 LLM via the Groq API to synthesize answers from retrieved context.

• **Graph Execution Module:** Orchestrates the workflow using LangGraph, managing state transitions between nodes.

• **HITL Module**: A specialized node that interrupts the automated flow to capture manual human expert input.

Data Structures

• **Document Representation**: Objects containing page_content (string) and metadata (source, page number).

• **Chunk Format**: Smaller string segments derived from documents, maintaining pointer references to the parent document.

• **Embedding Structure**: Numerical arrays (vectors) representing the semantic meaning of chunks.

• **Query-Response Schema**: Pydantic models defining the JSON structure for API requests (query) and responses (query, response, category).

• **State Object for Graph:** The AgentState TypedDict, which acts as the global shared memory for the graph, holding the query, context, response, and routing category

Workflow Design (LangGraph)

• **Nodes:**

- retrieve_node: Fetches context from ChromaDB.
- processing_node: Generates AI response and evaluates confidence.
- human_node: Pauses for manual agent input.

• **Edges:** Directed paths connecting nodes; including a conditional edge from processing_node to either human_node or END.

• **State:** The AgentState object that is updated at each node and passed to the next

Conditional Routing Logic

• **Answer Generation Criteria:** Triggered when the retriever finds relevant context and the LLM can form a confident answer.

• **Escalation Criteria:** Triggered if the query is identified as "Unique" or "Out of Scope".

• **Low Confidence**: If the LLM output explicitly contains "ESCALATE" or fails to find a clear answer in the text.

• **Missing Context:** Triggered automatically if the retrieve_node returns an empty context string.

• **Complex Query:** Detected when the intent requires reasoning beyond the provided PDF documentation

HITL Design

• **When Escalation is Triggered:** When the router function receives a state with category: escalate.

• **What Happens After Escalation**: The graph enters the human_node, printing the state to the terminal and pausing for a standard input.

• **How Human Response is Integrated:** The human's typed response is written to the response field of the AgentState, which is then returned as the final output.

API / Interface Design

• **Input Format:** text : {"query": "string"}.

• **Output Format:** Answer: {"query": "string", "response": "string", "category": "string"}

. • **Interaction Flow:** User submits question in UI and receives respons.

Error Handling

• **Missing Data:** Validation checks to ensure the data/ folder and PDF files exist before ingestion.

• **No Relevant Chunks Found:** The system is programmed to return a "context missing" status rather than allowing the LLM to guess.

• **LLM Failure:** retry request or display temporary error